

Notes

Chapter: Ecosystem and Energy flow

Which one of the following has the largest population in a food chain?

a. Producers b. Primary consumers c. Secondary consumers d. Decomposers

2. The second trophic level in a lake is _____

a. Phytoplankton b. Zooplankton c. Benthos d. Fishes

3. Secondary consumers are _____

a. Herbivores b. Producers c. Carnivores d. Autotrophs

4. What is the % of photosynthetically active radiation in the incident solar radiation?

a. 100% b. 50 % c. 1-5% d. 2-10%

6. Give the term used to express a community in its final stage of succession?

a. End community b. Final community c. Climax community d. Dark community

7. After landslide which other following type of succession occurs?

a. Primary b. Secondary c. Tertiary d. Climax

8. Which of the following is most often a limiting factor of the primary productivity in any ecosystem.

a. Carbon b. Nitrogen c. Phosphorus d. Sulphur

1. Tansley (1935) coined the term ecosystem.

2. Natural ecosystems do not require any human inputs, in other words they are self-sustainable

3. Artificial ecosystem e.g. a farm land, a fish tank or even a large pond used for rearing fish, require constant input in terms of energy or materials.

4. Biotic and abiotic components differ as the locations vary in space and time. The variation due to space results in spatial pattern

5. Vertical distribution of different species of plants and animals occupying different levels, is known as stratification.

6. Similar stratification is also observed in the open seas as epipelagic, meso-pelagic, bathy-pelagic and benthic zones

7. Horizontal distribution of plants and animals on land or in water, is called **zonation**.
8. The **energy flow** is unidirectional flow of energy from producers to consumers and finally dissipation and loss as heat.
9. **Productivity** refers to the rate of generation of biomass in an ecosystem
10. **Gross primary productivity** of an ecosystem is the rate of production of organic matter during photosynthesis
11. $GPP - R = NPP$
12. **Decomposition** is the break down of dead organic material and mineralization of the dead matter.
13. The process of **productivity** involves conversion of inorganic chemicals into organic material with the help of the radiant energy of the sun by the autotrophs and consumption of the autotrophs by heterotrophs
14. The **nutrient cycling** is the storage and transport of nutrients.
15. **Net primary productivity** is the available biomass for the consumption, to heterotrophs.
16. **Secondary productivity** is defined as the rate of formation of new organic matter by consumers.
17. Detritivores (e.g. earthworm) break down detritus into smaller fragments or particles. This process is called **fragmentation**
18. The humus is further degraded by some microbes and release of inorganic nutrients occurs by
19. Humification leads to accumulation of partially decomposed, a dark coloured, amorphous, colloidal organic substance called **humus** that is resistant to microbial action and undergoes decomposition at an extremely slow rate.
20. Of the total incident solar radiation, less than 50 % of it is **photosynthetically active radiation (PAR)**.
21. Directly or indirectly, all organisms are dependent for their food on producers. Hence there is **unidirectional flow of energy**
22. The autotrophs are called **producers**.
23. All animals directly or indirectly depend on plants for their food. They are hence called **consumers**.
24. If they feed directly on the plants, they are called **primary consumers**, and if the animals eat other animals which eat plants, they are called **secondary consumers**. Likewise, you could have **tertiary consumers** too.
25. The primary consumers are also known as **herbivores**.
26. The consumers that feed on these herbivores are **carnivores**.
27. The amount of energy available decreases at each successive trophic level. The number of trophic levels in any food chain is restricted as the transfer of energy follows '**10% Law**'
28. **food chains** do not exist in isolation, but are always interconnected to form food web that maintains the stability of an ecosystem.

29. The base of each pyramid represents the producers or the first trophic level while the apex represents tertiary or top level consumer.
30. The relative number of individuals per unit area at different trophic levels, constitutes the **number pyramid**.
31. Amount of accumulated energy per unit area, is **energy pyramid**.
32. These long-term storage places are known as "sinks".
33. Atmospheric inputs of **phosphorus** through rainfall are much smaller (measure) than carbon inputs.
34. Sudden influx of phosphorus in the form of agricultural runoff or industrial effluents rich in phosphate content, leads to **eutrophication** in water bodies.
35. **Succession** is a spatial pattern which occurs over the time. Time is a crucial factor.
36. The gradual and predictable change in the species composition of a given area is called **ecological succession**
37. The entire sequence of communities that successively change in a given area, constitute what is called **sere(s)**.
38. The individual transitional communities are termed **seral communities**.
39. Succession is hence a process that starts where no living organisms were present before - like on a newly formed volcanic island. This is called **primary succession**.
40. The species that invade a bare area, are called **pioneer species**
41. The **climax community** remains stable as long as the environment remains unchanged.
42. In **primary succession** of aquatic habitat, (steps 1 to 7 in fig. 14.13) the pioneers are the small phytoplankton.
43. In **secondary succession**, the species that invade depend on the condition of the soil, availability of water, the environment as also the seeds or other propagules present.
44. The products of ecosystem processes are named as **ecosystem services**, for example, healthy forest ecosystems purify air and water, mitigate droughts and floods.
45. **Supporting services** include services such as nutrient cycling, primary production, soil formation, habitat provision and pollination maintaining balance of ecosystem.
46. **Provisioning services** include food (including seafood), raw materials (including timber, skins, fuel wood), genetic resources (including crop improvement genes, and health care), water, medicinal resources (including test and assay organisms) and ornamental resources (furs, feathers, ivory, orchids, butterflies, etc).
47. **Regulating services** include Carbon sequestration, Predation regulates prey populations, Waste decomposition and detoxification, Purification of water and air, and pest control.
48. **Cultural services** include cultural, spiritual and historical, recreational experiences, science and education, and Therapeutics (including animal assisted therapy)
49. **Fixation of atmospheric CO₂** and **release of O₂** are the most important services provided by an ecosystem.
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Q. 2 Very short answer question. (1 mark questions)

1. Give an example of ecosystem which shows inverted pyramid of numbers.

Ans: Pyramid of numbers can also be sometimes inverted. e.g. In a small ecosystem, if we plot the number of insects on a single tree, smaller birds feeding on insects, and parasites on those birds, we get an inverted pyramid.

2. Give an example of ecosystem which shows inverted pyramid of biomass.

Ans: Oceanic ecosystem show inverted biomass pyramid.

3. Which mineral acts as limiting factor for productivity in an aquatic ecosystem

Ans: Phosphorus acts as limiting factor for productivity in an aquatic ecosystem

. 4. Name the reservoir and sink of carbon in carbon cycle.

Ans: Carbon that is a part of rocks and fossil fuels like oil, coal and natural gas, has been away from the rest of the carbon cycle for a long time. These long-term storage places are known as “sinks”.

Q. 3 Short answer questions. (2/3 marks question)

1. Distinguish between upright and inverted pyramid of biomass

2. Distinguish between Food chain and Food web.

Upright pyramid

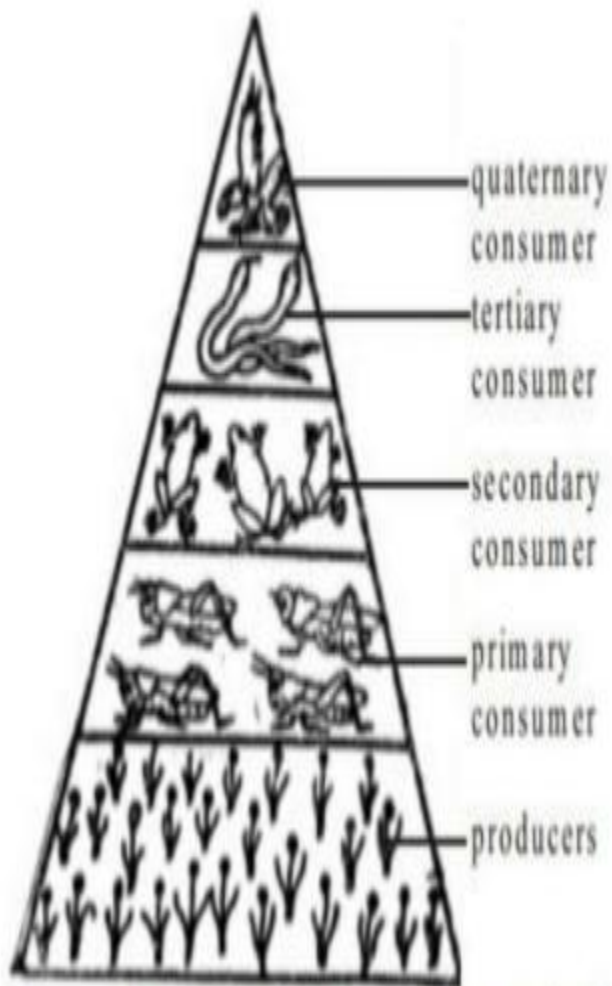
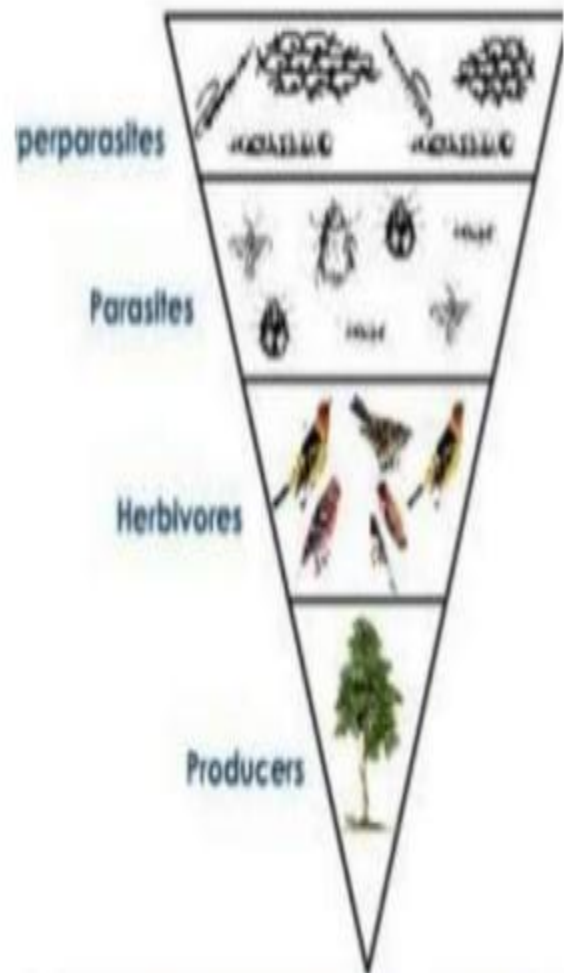


Fig. Pyramid of numbers in a grassland

Inverted pyramid



Inverted pyramid of number

Upright pyramid	Inverted pyramid
In this type of pyramid the number and biomass of organisms at producer level is high, which keeps on decreasing at each trophic level.	In this type of pyramid the number and biomass of organisms at producer level is lowest and keeps on increasing at each trophic level.
The base bar comprises of producers in large number.	The base bar comprises of producers in small number.
Energy pyramid is always upright.	Pyramid of number and pyramid of biomass can be inverted.

(ii) Differences between food chain and food web are:

Food chain	Food web
It is a linear sequence of organisms.	It is an interconnection of many food chains.
Members present at higher trophic level feeds on single type of organisms.	One organism has alternate food sources.
The energy flow can be easily calculated.	Energy flow is difficult to calculate.
Competition is limited only to the members of same trophic level.	Competition is amongst members of same and different trophic levels.

Q 4. Long answer questions. (4 marks question)

1. Define ecological pyramids and describe with examples, pyramids of number and biomass.

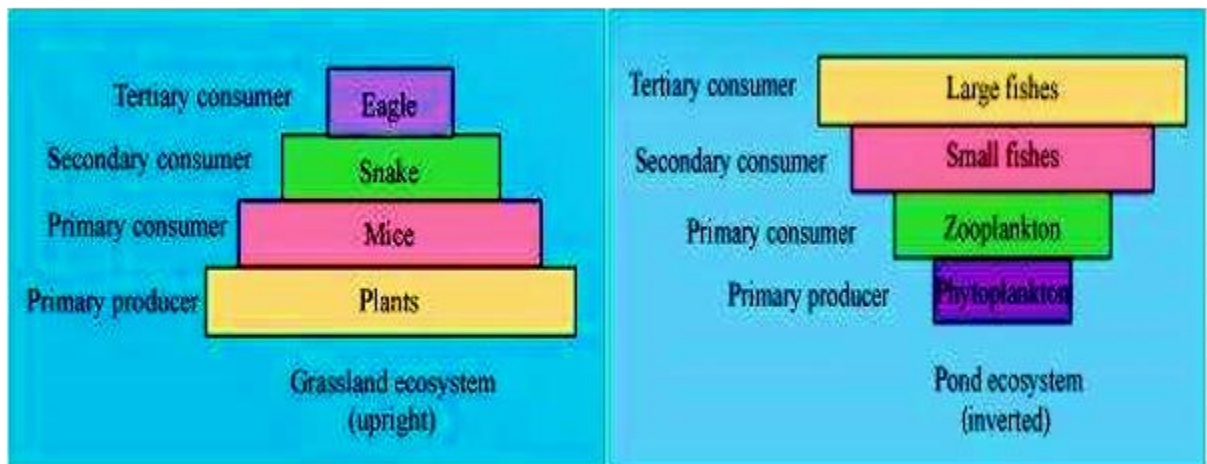
Ans: Ecological Pyramid:

1. Ecological pyramid is a graphic representation of the relationship between the organisms of various successive trophic levels with respect to energy, biomass and number.
2. The base of each pyramid represents the producers or the first trophic level while the apex represents tertiary or top level consumer.
3. The three ecological pyramids which are usually studied are : **Pyramid of biomass**, **Pyramid of numbers** and **Pyramid of energy**.

A) Pyramid of Number:

1. **Pyramid of numbers:** It is a graphical representation of the number of individuals present at each trophic level in a food chain of an ecosystem. The pyramid of numbers can be upright or inverted depending on the number of producers. For example, in a grassland ecosystem, the pyramid of numbers is upright. In this type of a food chain, the number of producers (plants) is followed by the number of

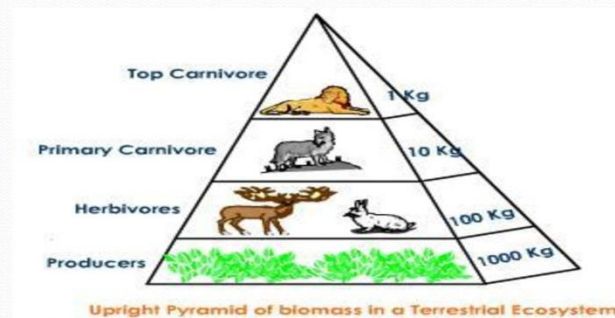
herbivores (mice), which in turn is followed by the number of secondary consumers (snakes) and tertiary carnivores (eagles). Hence, the number of individuals at the producer level will be the maximum, while the number of individuals present at top carnivores will be least.



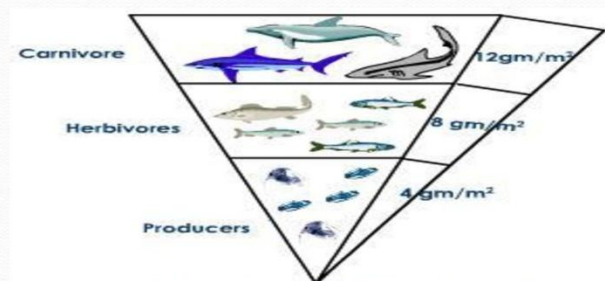
2. Pyramid of Biomass

A pyramid of biomass is a graphical representation of the total amount of living matter present at each trophic level of an ecosystem. It can be upright or inverted. It is upright in grasslands and forest ecosystems as the amount of biomass present at the producer level is higher than at the top carnivore level. The pyramid of biomass is inverted in a pond ecosystem as the biomass of fishes far exceeds the biomass of zooplankton (upon which they feed).

Pyramids of Biomass cont.



Upright



Inverted

2. What is primary productivity? Give brief description of factors that affect primary productivity.

- **Ans:** Productivity refers to the rate of generation of biomass in an ecosystem.
- It is expressed in units of mass per unit surface (or volume) per unit time, for instance grams per square metre per day ($\text{g/m}^2/\text{day}$).
- The mass unit may relate to dry matter or to the mass of carbon generated.

Factors affecting Primary Productivity

- Primary productivity (GPP) depends on the plant species inhabiting a particular area.
- It also depends on a variety of environmental factors, availability of nutrients and photosynthetic capacity of plants.
- Therefore, it varies in different types of ecosystems

3. Define decomposition and describe the processes and products of decomposition

Ans:Decomposition:Decomposers break down complex organic matter into inorganic substances like carbon dioxide, water and nutrients, and the process is called **decomposition**.

Decomposition process:

The important steps in the process of decomposition are:-

- 1) fragmentation,
 - 2) leaching,
 - 3) catabolism,
 - 4) humification and
 - 5) mineralization.
- 1) **Fragmentation:**Detritivores (e.g. earthworm) break down detritus into smaller fragments or particles. This process is called **fragmentation**.
 - 2) **Leaching:**-By the process of **leaching**, water soluble inorganic nutrients go down (percolate) into the soil horizon and get precipitated as unavailable salts.
 - 3) **Catabolism:** Bacterial and fungal enzymes degrade detritus into simpler inorganic substances. This process is called as **catabolism**. It is important to note that all the above steps in decomposition operate simultaneously on the detritus.
 - 4) **Humification:** Humification leads to accumulation of partially decomposed, a dark coloured, amorphous, colloidal organic substance called **humus** that is resistant to microbial action and undergoes decomposition at an extremely slow rate. Humus formation changes soil texture and increases water holding capacity of soil.
 - 5) **Mineralisation:**-Being colloidal in nature humus serves as a reservoir of nutrients. The humus is further degraded by some microbes and release of inorganic nutrients occurs by the process known as **mineralization**

Factors that do not favour Decomposition

Decomposition as a process requires oxygen. Temperature and soil moisture are the most important factors that regulate decomposition indirectly to help soil microbes. Warm and moist environment favours decomposition whereas low temperature and anaerobic conditions inhibit decomposition

Products of Decomposition

products are released: **carbon dioxide** (CO₂), energy, water, plant nutrients and resynthesized organic carbon compounds

. Successive decomposition of dead material and modified organic matter results in the formation of a more complex organic matter called humus.

4. Write important features of a sedimentary cycle in an ecosystem.

Ans:-Sedimentary cycle is the cycle of mineral found in the earth crust e.g., phosphorus, sulphur etc. Important features of a sedimentary in an ecosystem are:

- 1) It is the major reservoir for nutrients elements in the lithosphere and the elements are released by weathering.
- 2)E.g. phosphorous cycle, sulphur cycle, iodine cycle.
- 3)The sedimentary cycle is very slow than gaseous cycle and takes more time.
- 4)It is a less perfect cycle as elements get locked in earth crust for the longer time.

5. Discribe carbon cycle and add a note on the impact of human activities on carbon cycle

Ans:Carbon Cycle Steps

I)Carbon in the Atmosphere

1. To become part of the carbon cycle, carbon atoms start out in a gaseous form. Carbon dioxide gas – CO₂ – can be produced by inorganic processes, or by the metabolisms of living things.
- 2. Today, carbon is also released into the atmosphere through the activities of living things, such as the exhalations of animals, the actions of **decomposer** organisms, and the burning of wood and fossil fuels by humans.**
3. However carbon dioxide gets into the atmosphere, CO₂ gas is the starting point of the carbon cycle. The next step is...

II) Producers Absorb Carbon

- 1.“**Producers**” – organisms that produce food from sunlight, such as plants – absorb carbon dioxide from the atmosphere and use it to build sugars, lipids, proteins, and other essential building blocks of life.
2. For plants, CO₂ is absorbed through pores in their leaves called “stomata.”
- 3. Carbon dioxide enters the **plant** through the stomata and is incorporated into containing carbon compounds with the help of energy from sunlight.**
4. Plants and other **producer** organisms such as cyanobacteria are crucial to life on Earth because they can turn atmospheric carbon into living matter.

III)Producers are Eaten

1. “**Consumers**” are organisms that eat other living things.
- 2. Animals are the most visible type of **consumer** in our ecosystems, though many types of microbes also fall into this category.**
3. Consumers incorporate carbon compounds from plants and other food sources when they eat them.
4. They use some of these carbon compounds from food to build their own bodies – but much of the food they eat is broken down to release energy, in a process that is almost the reverse of what producers do.
5. These are ultimately released into the atmosphere in the form of CO₂.
6. But, what about the carbon compounds that don’t get eaten, or broken down by animals?

IV) Decomposers Release Carbon

1. Plants and animals that die without being eaten by other animals are broken down by other organisms, called “decomposers.”
- 2. Decomposers include many **bacteria** and some **fungi**.**

3. They usually only break down matter that is already dead, rather than catching and eating a living animal or plant.
4. Just like animals, decomposers break down the chemical bonds in their food molecules. They create many chemical products, including in some cases CO₂.

V) Human Activities

1. Recently, humans have made some big changes to the Earth's carbon cycle. By burning huge amounts of fossil fuels and cutting down roughly half of the Earth's forests, humans have decreased the Earth's ability to take carbon out of the atmosphere, while releasing large amounts of carbon into the atmosphere that had been stored in solid form as plant matter and fossil fuels.
2. **This means more carbon dioxide in Earth's atmosphere – which is particularly dangerous since carbon dioxide is a “greenhouse gas” that plays a role in regulating the Earth's temperature and weather patterns**

.Application Based Questions

1. **Why is zonation more pronounced at the edges of habitat?**
Ans: In **ecology**, **edge effects** are changes in **population** or **community** structures that occur at the boundary of two or more **habitats**.^[1] Areas with small **habitat fragments** exhibit especially pronounced edge. As the edge effects increase, the boundary habitat allows for greater **biodiversity**.
2. **What could be the reason for the low productivity of ocean?**
Ans: The biomass in the **oceans** is **low**. - The **cause of** this is lack of enough sunlight to enable high rates of photosynthesis which **would** increase **productivity**
3. **What is a food chain?**
Ans: The term **food chain** refers to the sequence of events in an ecosystem, where one organism eats another and then is eaten by another organism
4. **. What are trophic levels in a food chain?**
Ans: The **trophic level** of an organism is the number of steps it is from the start of the **chain**. A **food chain** starts at **trophic level 1** with primary producers such as plants, can move to herbivores at **level 2**, carnivores at **level 3** or higher, and typically finish with apex predators at **level 4** or **5**
5. **Is there any presence of living organisms in the perpetual darkness of deep oceanic trenches?**
Ans: The abyssal zone remains in **perpetual darkness** at a depth of 4,000 to 6,000 metres (13,000 to 20,000 ft). The only **organisms** that inhabit this zone are chemotrophs and predators that can withstand immense pressures, sometimes as high as 76 megapascals (750 atm; 11,000 psi)
6. **In absence of solar radiation, what is their source of energy?**
Ans: Even without sunlight, most life in **the deep sea** is still dependent on **the sun**. **The** dead organic matter on which these animals and bacteria feed ultimately comes from

phytoplankton and other plants and organisms that harness sunlight to create **energy** in a process called photosynthesis.

7. **Which organisms do serve as producers in the food chains of deep oceans?**

Ans: The **producers** are the phytoplankton (Greek phyton = plant) and the consumers, who eat the phytoplankton, are the zooplankton (Greek zoon = animal).

8. **What could be the connecting points between the GFC and DFC?**

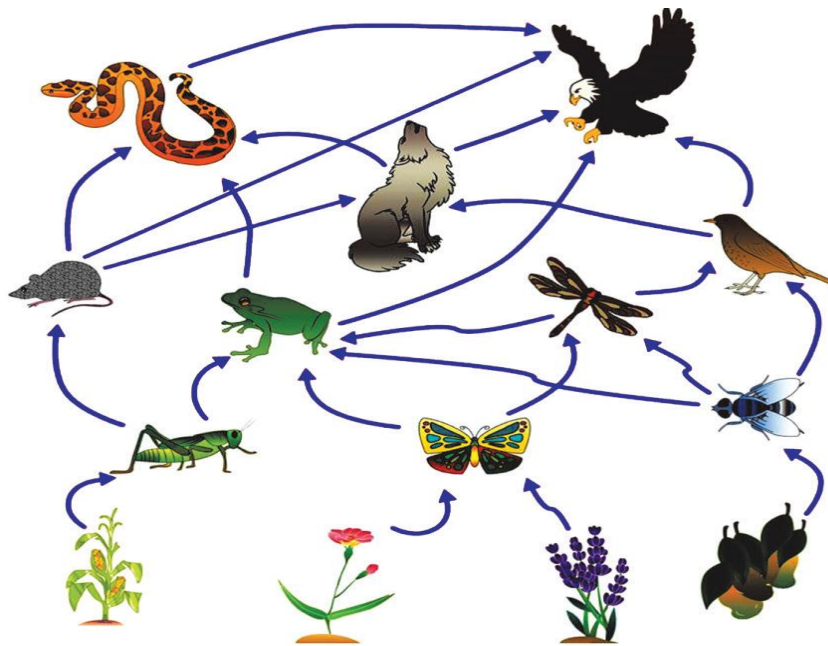
Ans:In this **food chain**, energy comes from organic matter (or **detritus**) generated in trophic levels of the **grazing food chain**. ... It begins with **detritus** such as dead bodies of animals or fallen leaves, which are then eaten by decomposers or detritivores. These detritivores are in turn consumed by their predators

9. **How will you classify man as carnivore (primary/ secondary) or omnivore? Why?**

Ans: **Carnivores** are living **beings** that eat other animals as food for their survival, example - tiger, lion. ... **Omnivores** are those living **beings** that depend on both plants and animals as food for their survival. As **humans** eat both vegetables as well as animals(in form of meat), they are considered to be **omnivores**

10. **How many trophic levels human beings function in a food chain?**

Ans: Human beings function at all the three **trophic levels** in the **food chain**: Primary trophic levelSecondary trophic levelTertiary trophic level



11. _____

12. **From the given food web diagram, give the trophic levels where the eagle is present**

Ans: Eagle is present as third level carnivore.

13. **What is the maximum number of trophic levels in a food chain?**

Ans: Generally, there are a **maximum** of four **trophic levels**. Many consumers feed at more than one **trophic level**.

14. How would you explain inverted pyramid of biomass in oceanic ecosystem?

Ans: The **pyramid of biomass** in sea is **inverted** because **biomass** of fishes exceeds that of phytoplankton. In this case the **biomass** of trophic level depends on the reproductive potential and the longevity of the member. **Marine** environments have **inverted biomass pyramids**

15. What are the long term and short-term effects of eutrophication in water bodies?

Ans: Eutrophication can have serious effects

Short Term Effect: algal blooms that block light from getting into the **water** and harm the plants and animals that need it. I

Long term effect there's enough overgrowth of algae, it can prevent oxygen from getting into the **water**, making it hypoxic and creating a dead zone where no organisms can survive.

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